

Diagram illustrating the I/O connections for the Raspberry Pi 4B, showing 16 green circular connectors (labeled 1 through 16) and their corresponding Read Switches (1 through 8) connected to the Raspberry Pi 4B (To_RPiB) and the Raspberry Pi 4B (To_RPi).

Connector Label	Connector Number	Read Switch	To_RPiB	To_RPi
Kitchen Bench	1	Read Switch 1	To_RPiB	Read1_input
Kitchen Panel	2	Read Switch 2	To_RPiB	Read2_input
Storage Panel	3	Read Switch 3	To_RPiB	Read3_input
Rear Drawer	4	Read Switch 4	To_RPiB	Read4_input
Rooftop Tent	5	Read Switch 5	To_RPiB	Read5_input
Spare/Future 1	6	Read Switch 6	To_RPiB	Read6_input
Spare/Future 2	7	Read Switch 7	To_RPiB	Read7_input
Spare/Future 3	8	Read Switch 8	To_RPiB	Read8_input
Read Switches	9-16	Read Switch 9-16	To_RPiB	Read9_input - Read16_input

A red arrow points from the bottom of the connector list to a red circle labeled GND.

Ambient Air Temp

Temp_sensor_inputD Temp1_sensor_input

The diagram illustrates three sensor inputs connected to a pin header:

- Water Level Sensor:** A red box labeled "Sensor_input0" is connected to the "Water_sensor_input" pin.
- Analog Input 1:** A red box labeled "Sensor_input0" is connected to the "Analog_input1" pin.
- Analog Input 2:** A red box labeled "Sensor_input0" is connected to the "Analog_input2" pin.

Below each sensor input box, the file path `File: analog_inputs.kicad_sch` is listed.

[illegible][illegible]

The diagram illustrates the pin connections for the RPL_W5281X_output2 module on a Raspberry Pi HAT. The HAT is shown with its 40 pins, and the connections are as follows:

- Power and Ground:**
 - Pin 1: +5V
 - Pin 2: GND
 - Pin 3: +5V
 - Pin 4: GND
 - Pin 5: +5V
 - Pin 6: GND
 - Pin 7: +5V
 - Pin 8: GND
 - Pin 9: +5V
 - Pin 10: GND
 - Pin 11: +5V
 - Pin 12: GND
 - Pin 13: +5V
 - Pin 14: GND
 - Pin 15: +5V
 - Pin 16: GND
 - Pin 17: +5V
 - Pin 18: GND
 - Pin 19: +5V
 - Pin 20: GND
 - Pin 21: +5V
 - Pin 22: GND
 - Pin 23: +5V
 - Pin 24: GND
 - Pin 25: +5V
 - Pin 26: GND
 - Pin 27: +5V
 - Pin 28: GND
 - Pin 29: +5V
 - Pin 30: GND
 - Pin 31: +5V
 - Pin 32: GND
 - Pin 33: +5V
 - Pin 34: GND
 - Pin 35: +5V
 - Pin 36: GND
 - Pin 37: +5V
 - Pin 38: GND
 - Pin 39: +5V
 - Pin 40: GND
- GPIO Connections:**
 - GPIO2 (SDA) to R1 (Temp1_sensor_input)
 - GPIO3 (SCL) to R2 (Temp1_sensor_input)
 - GPIO4 (GCLK0) to R3 (Temp1_sensor_input)
 - GPIO17 to R4 (Temp1_sensor_input)
 - GPIO18 (CLM_CLK) to R5 (Temp1_sensor_input)
 - GPIO23 to R6 (Temp1_sensor_input)
 - GPIO24 to R7 (Temp1_sensor_input)
 - GPIO26 to R8 (Temp1_sensor_input)
 - GPIO27 to R9 (Temp1_sensor_input)
 - GPIO28 to R10 (Temp1_sensor_input)
 - GPIO29 to R11 (Temp1_sensor_input)
 - GPIO30 to R12 (Temp1_sensor_input)
 - GPIO31 to R13 (Temp1_sensor_input)
 - GPIO32 to R14 (Temp1_sensor_input)
 - GPIO33 to R15 (Temp1_sensor_input)
 - GPIO34 to R16 (Temp1_sensor_input)
 - GPIO35 to R17 (Temp1_sensor_input)
 - GPIO36 to R18 (Temp1_sensor_input)
 - GPIO37 to R19 (Temp1_sensor_input)
 - GPIO38 to R20 (Temp1_sensor_input)
 - GPIO39 to R21 (Temp1_sensor_input)
 - GPIO40 to R22 (Temp1_sensor_input)
- Other Connections:**
 - GPIO1 (Reserved for BT) to R23 (Temp1_sensor_input)
 - GPIO5 (Reserved for BT) to R24 (Temp1_sensor_input)
 - GPIO10 (Reserved for BT) to R25 (Temp1_sensor_input)
 - GPIO11 (Reserved for BT) to R26 (Temp1_sensor_input)
 - GPIO12 (Reserved for BT) to R27 (Temp1_sensor_input)
 - GPIO13 (Reserved for BT) to R28 (Temp1_sensor_input)
 - GPIO14 (Reserved for BT) to R29 (Temp1_sensor_input)
 - GPIO15 (Reserved for BT) to R30 (Temp1_sensor_input)
 - GPIO16 (Reserved for BT) to R31 (Temp1_sensor_input)
 - GPIO19 (Reserved for BT) to R32 (Temp1_sensor_input)
 - GPIO20 (Reserved for BT) to R33 (Temp1_sensor_input)
 - GPIO21 (Reserved for BT) to R34 (Temp1_sensor_input)
 - GPIO22 (Reserved for BT) to R35 (Temp1_sensor_input)
 - GPIO25 (Reserved for BT) to R36 (Temp1_sensor_input)
 - GPIO26 (Reserved for BT) to R37 (Temp1_sensor_input)
 - GPIO27 (Reserved for BT) to R38 (Temp1_sensor_input)
 - GPIO28 (Reserved for BT) to R39 (Temp1_sensor_input)
 - GPIO29 (Reserved for BT) to R40 (Temp1_sensor_input)
 - GPIO30 (Reserved for BT) to R41 (Temp1_sensor_input)
 - GPIO31 (Reserved for BT) to R42 (Temp1_sensor_input)
 - GPIO32 (Reserved for BT) to R43 (Temp1_sensor_input)
 - GPIO33 (Reserved for BT) to R44 (Temp1_sensor_input)
 - GPIO34 (Reserved for BT) to R45 (Temp1_sensor_input)
 - GPIO35 (Reserved for BT) to R46 (Temp1_sensor_input)
 - GPIO36 (Reserved for BT) to R47 (Temp1_sensor_input)
 - GPIO37 (Reserved for BT) to R48 (Temp1_sensor_input)
 - GPIO38 (Reserved for BT) to R49 (Temp1_sensor_input)
 - GPIO39 (Reserved for BT) to R50 (Temp1_sensor_input)
 - GPIO40 (Reserved for BT) to R51 (Temp1_sensor_input)

The diagram also shows connections for the RPL_W5281X_output2 module, including Read1_Input, Read2_Input, Read3_Input, ESP32_2_rx, Read4_Input, GPS_tx, Read5_Input, Read6_Input, Read7_Input, and Read8_Input. The module is connected to the HAT via a ribbon cable. The HAT is also connected to a PWR_FLAG, +3V3, and GND.

ESP32-S3-WROOM-1 No. 1

Water_sensor_inputD — Water_sensor_inputD

NC — NC

LED1_outputD — Led1_output

LED2_outputD — Led2_output

LED3_outputD — Led3_output

LED4_outputD — Led4_output

LED5_outputD — Led5_output

LED6_outputD — Led6_output

LED13_outputD — Led13_output

LED14_outputD — Led14_output

File: esp32.kicad_sch

ESP32-S3-WROOM-1 No. 2

Input 1D — Analog_input1

Input 2D — Analog_input2

LED7_outputD — Led7_output

LED8_outputD — Led8_output

LED9_outputD — Led9_output

LED10_outputD — Led10_output

LED11_outputD — Led11_output

LED12_outputD — Led12_output

LED15_outputD — Led15_output

LED16_outputD — Led16_output

File: esp32.kicad_sch

LED Channel 1 White Led1_output → Pwm_in Pwm_out

LED Channel 2 Red Led2_output → Pwm_in Pwm_out

LED Channel 3 Green Led3_output → Pwm_in Pwm_out

LED Channel 4 White Led4_output → Pwm_in Pwm_out

LED Power 1 12Vdc_out 12Vdc_in GND

LED Channel 5 White Led5_output → Pwm_in Pwm_out

LED Channel 6 White Led6_output → Pwm_in Pwm_out

LED Channel 7 White Led7_output → Pwm_in Pwm_out

LED Channel 8 White Led8_output → Pwm_in Pwm_out

LED Power 2 12Vdc_out 12Vdc_in GND

LED Channel 9 White Led9_output → Pwm_in Pwm_out

LED Channel 10 Red Led10_output → Pwm_in Pwm_out

LED Channel 11 Green Led11_output → Pwm_in Pwm_out

LED Channel 12 White Led12_output → Pwm_in Pwm_out

LED Power 3 12Vdc_out 12Vdc_in GND

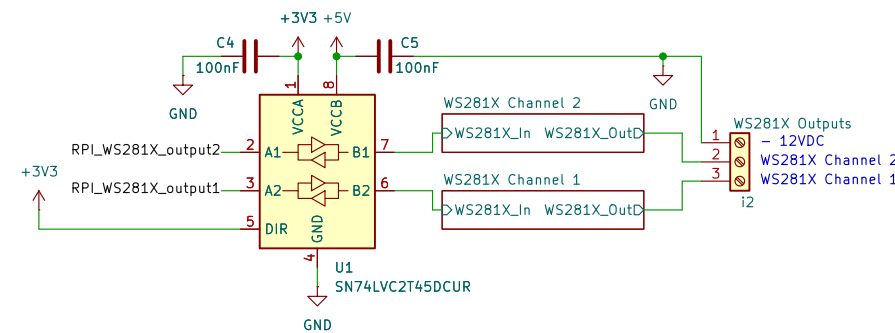
LED Channel 13 White Led13_output → Pwm_in Pwm_out

LED Channel 14 White Led14_output → Pwm_in Pwm_out

LED Channel 15 White Led15_output → Pwm_in Pwm_out

LED Channel 16 White Led16_output → Pwm_in Pwm_out

LED Power 4 12Vdc_out 12Vdc_in GND

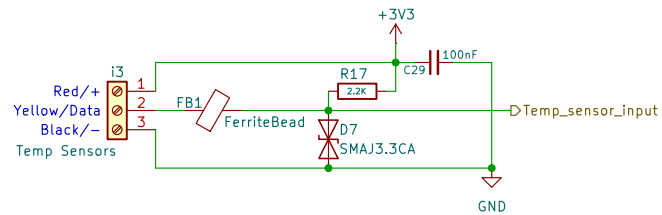


The diagram illustrates a multi-channel relay system. On the left, four input labels are listed vertically: Relay1_output, Relay2_output, Relay3_output, and Relay4_output. Each label is connected by a horizontal line to a corresponding 'Relay_output' label on the right. These four 'Relay_output' labels are grouped together within a single large red rectangular box. Above this box, the text 'Relay Channel 1' is written. Above the second 'Relay_output' label, the text 'Relay Channel 2' is written. Above the third 'Relay_output' label, the text 'Relay Channel 3' is written. Above the fourth 'Relay_output' label, the text 'Relay Channel 4' is written. The entire diagram is set against a light blue background.

1. Power circuit provides 5V
2. RPI Provides 3.3V
3. 1-Wire Bus Notes:
Board has a 2.2K pull-up on the data line.
Put a 220R resistor on the data pin of each temp sensor but ensure it's not in series with the bus.
Put a 100nF ceramic cap across VCC and GND at each temp sensor.
4. WS281X RPI Notes:
Put a 220uF electrolytic cap between 12V and GND at the start of each LED strip
5. UARTS:
UART1 / GPIO14/15 / Reserved for Bluetooth
UART2 / GPIO00/01 / GPS
UART3 / GPIO04/05 / ESP32 No. 1
UART4 / GPIO08/09 / ESP32 No. 2
6. 12V LED Strip Notes (not required but good practice):
Put a 220uF 25V electrolytic cap across 12V and GND at each LED strip.
Put a 100nF 25V ceramic cap across 12V and GND at each LED strip.

Scale: NTS
 Drawn By: Joel Goudswaard
Singularity Automation Pty Ltd
 Sheet: /
 File: SCCS-RPL.kicad_sch
Title: Singularity Camper Control System (SCCS)

Size: A2	Date: 2026-08-10	Rev: 1.1
KiCad E.D.A. 10.0.4		Id: 1/41



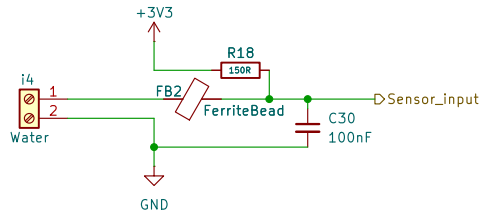
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Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 2/41



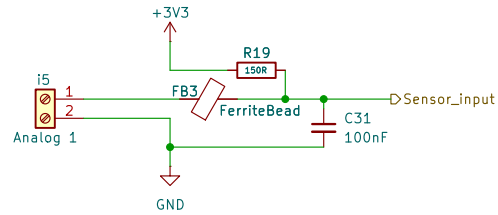
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Title:

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KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 3/41



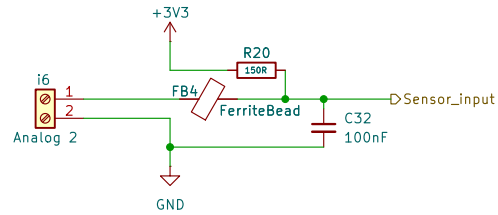
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Title:

Size: A5
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Date:

Rev:
Id: 4/41



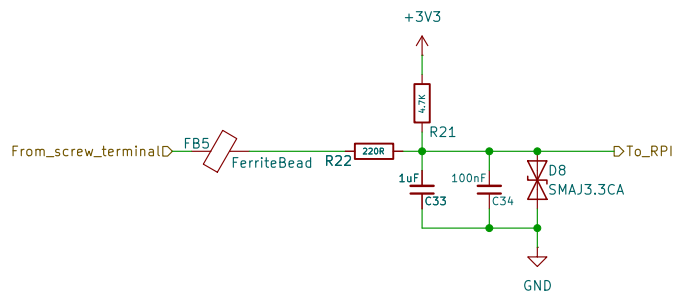
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Title:

Size: A5
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Date:

Rev:
Id: 5/41



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File: reed_input.kicad_sch

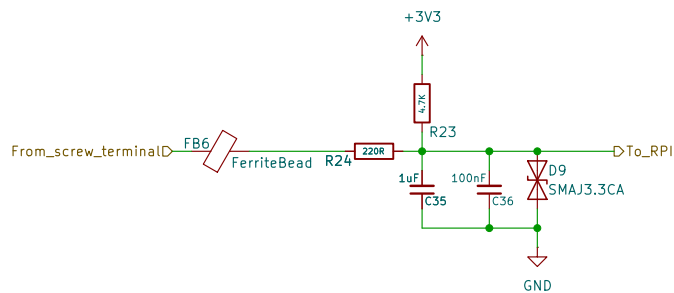
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Date:

Rev:

Id: 6/41



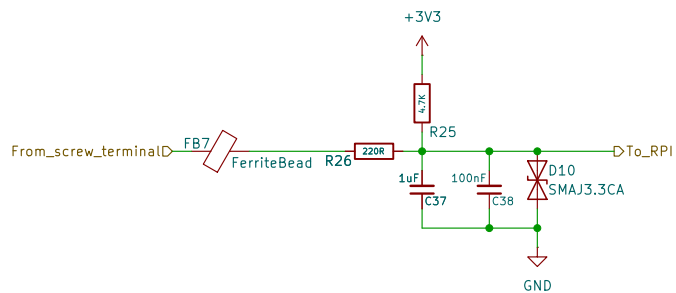
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Title:

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Date:

Rev:
Id: 7/41



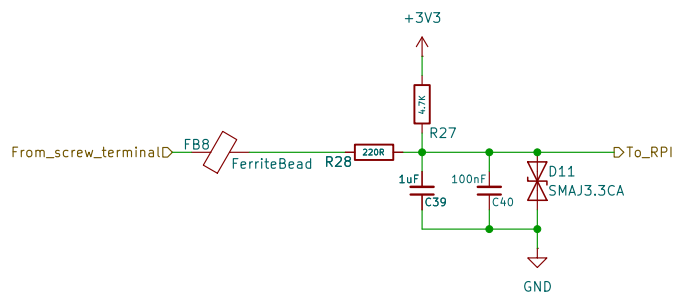
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Title:

Size: A5
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Date:

Rev:
Id: 8/41



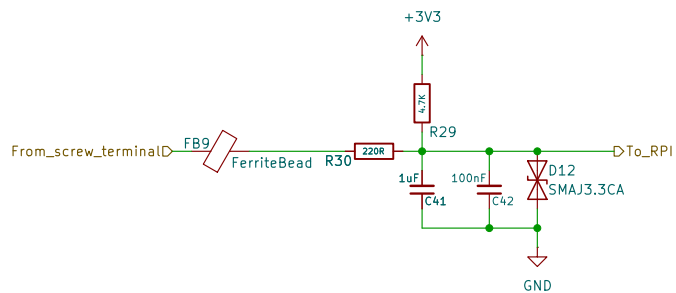
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Title:

Size: A5
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Date:

Rev:
Id: 9/41



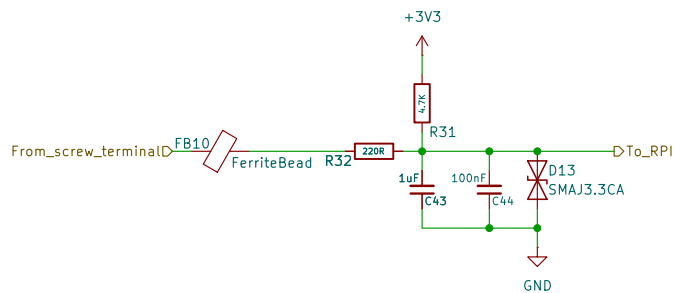
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Title:

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Date:

Rev:
Id: 10/41



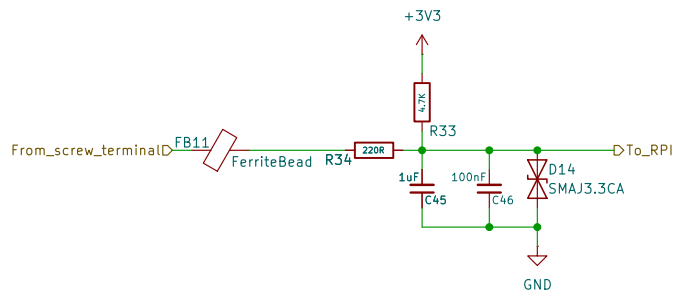
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Title:

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Date:

Rev:
Id: 11/41



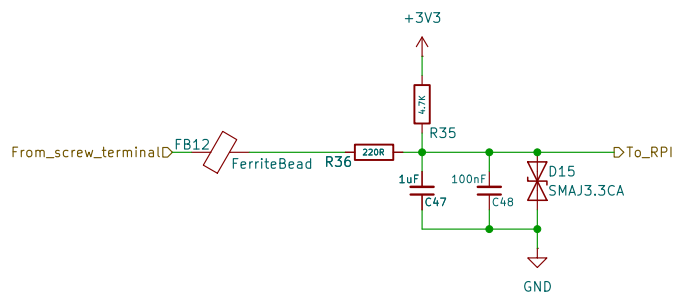
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Title:

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Date:

Rev:
Id: 12/41



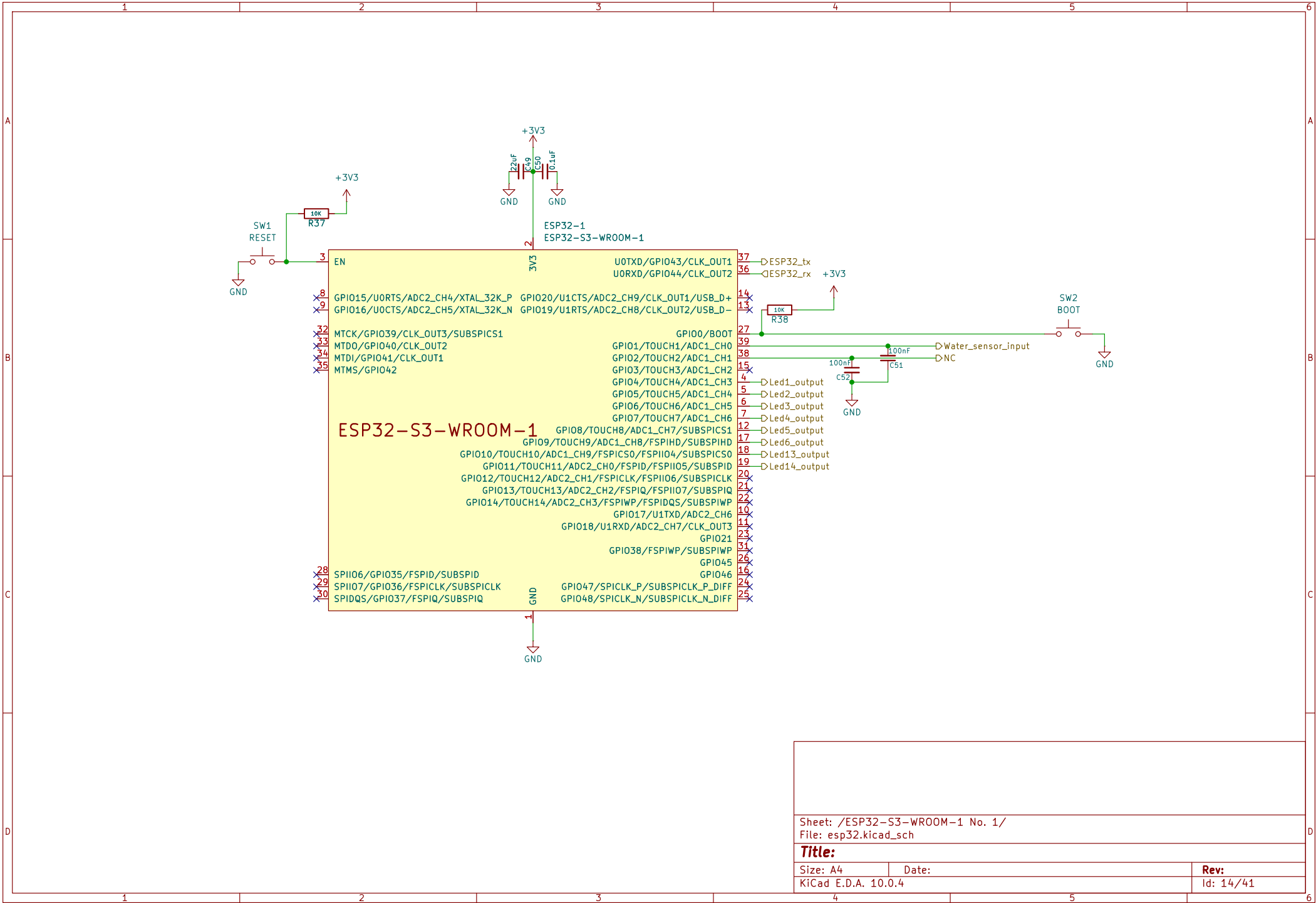
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Title:

Size: A5
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Date:

Rev:
Id: 13/41



Sheet: /ESP32-S3-WROOM-1 No. 1/
File: esp32.kicad_sch

Title:

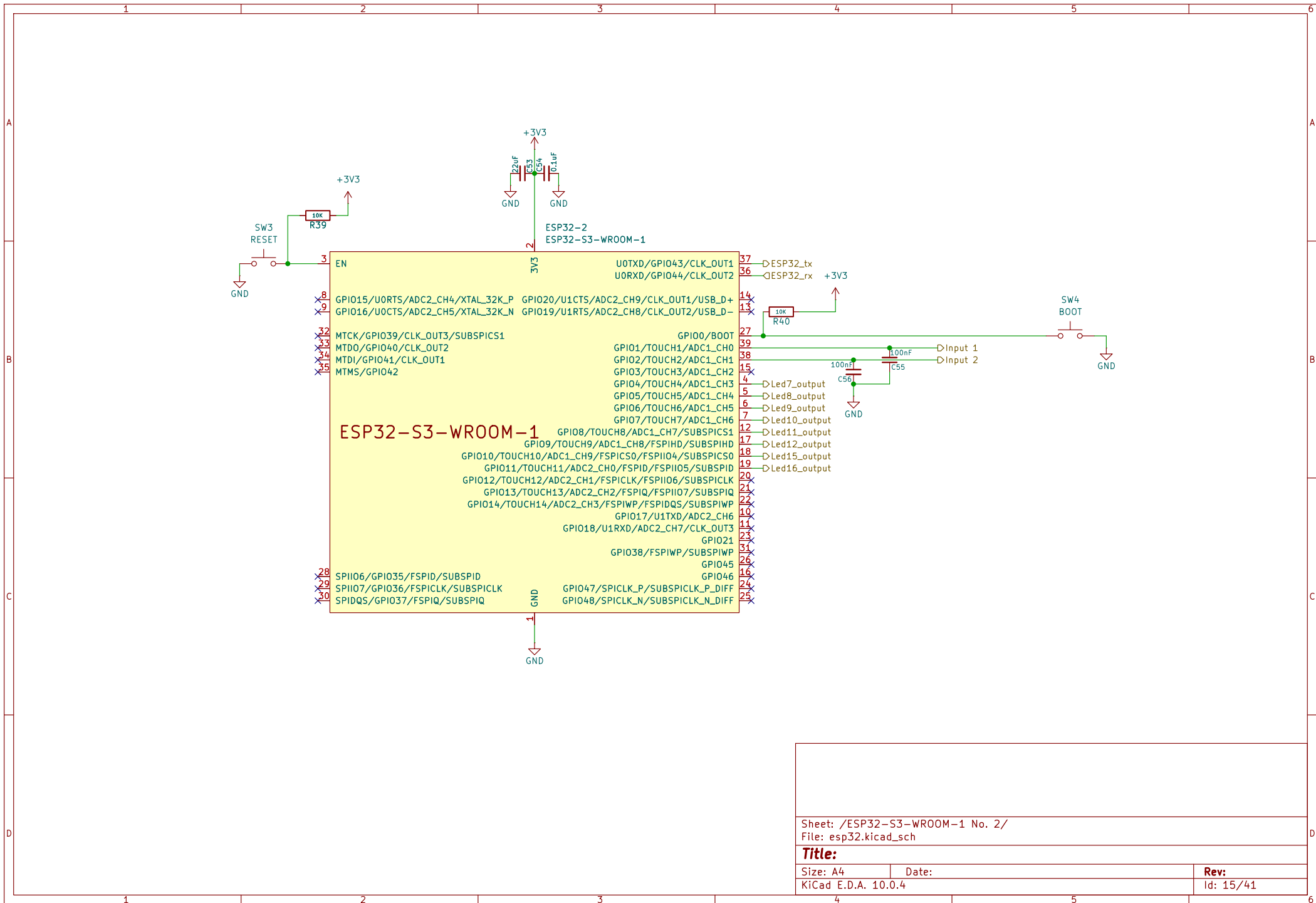
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Rev:

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Id: 14/41



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File: esp32.kicad_sch

Title:

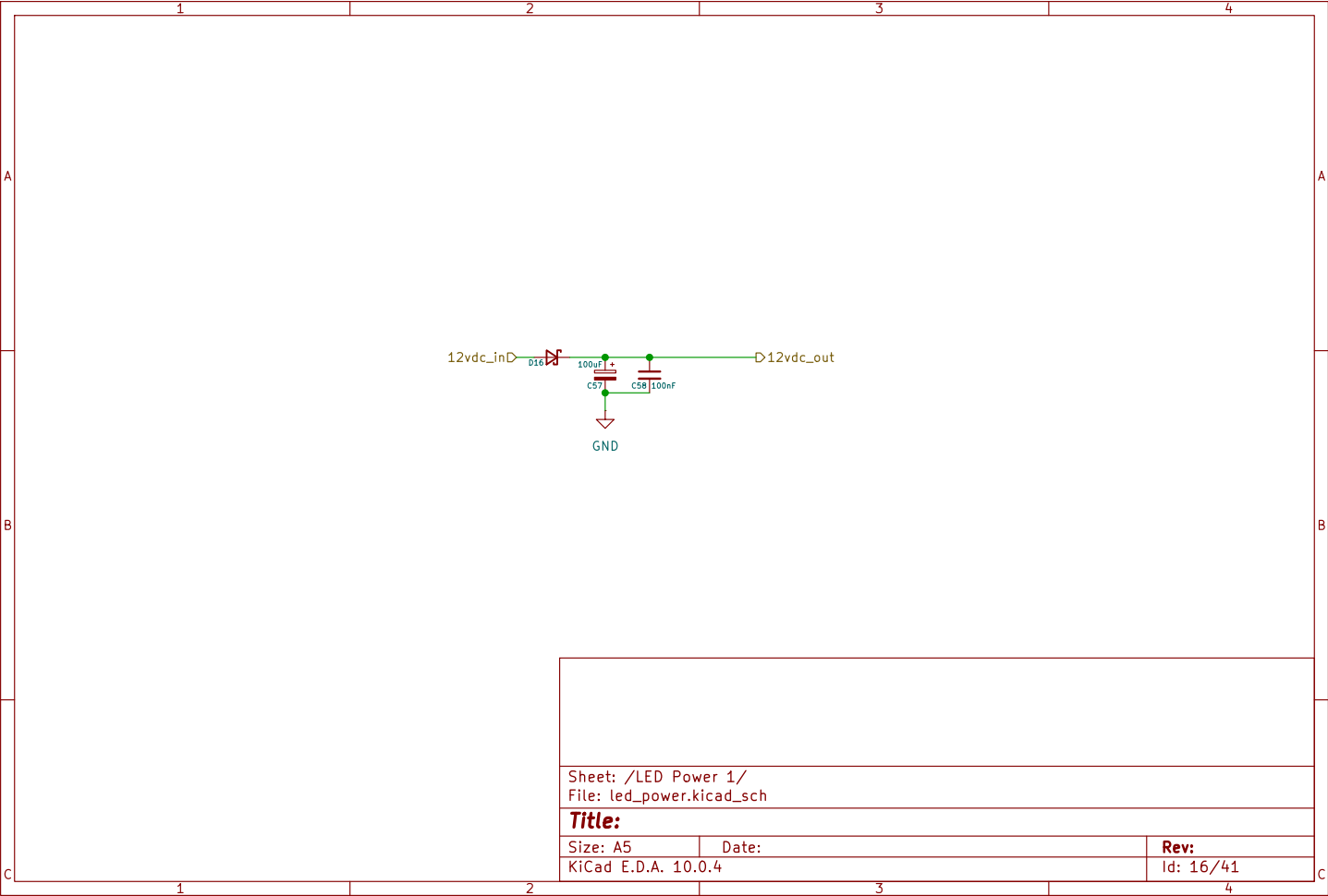
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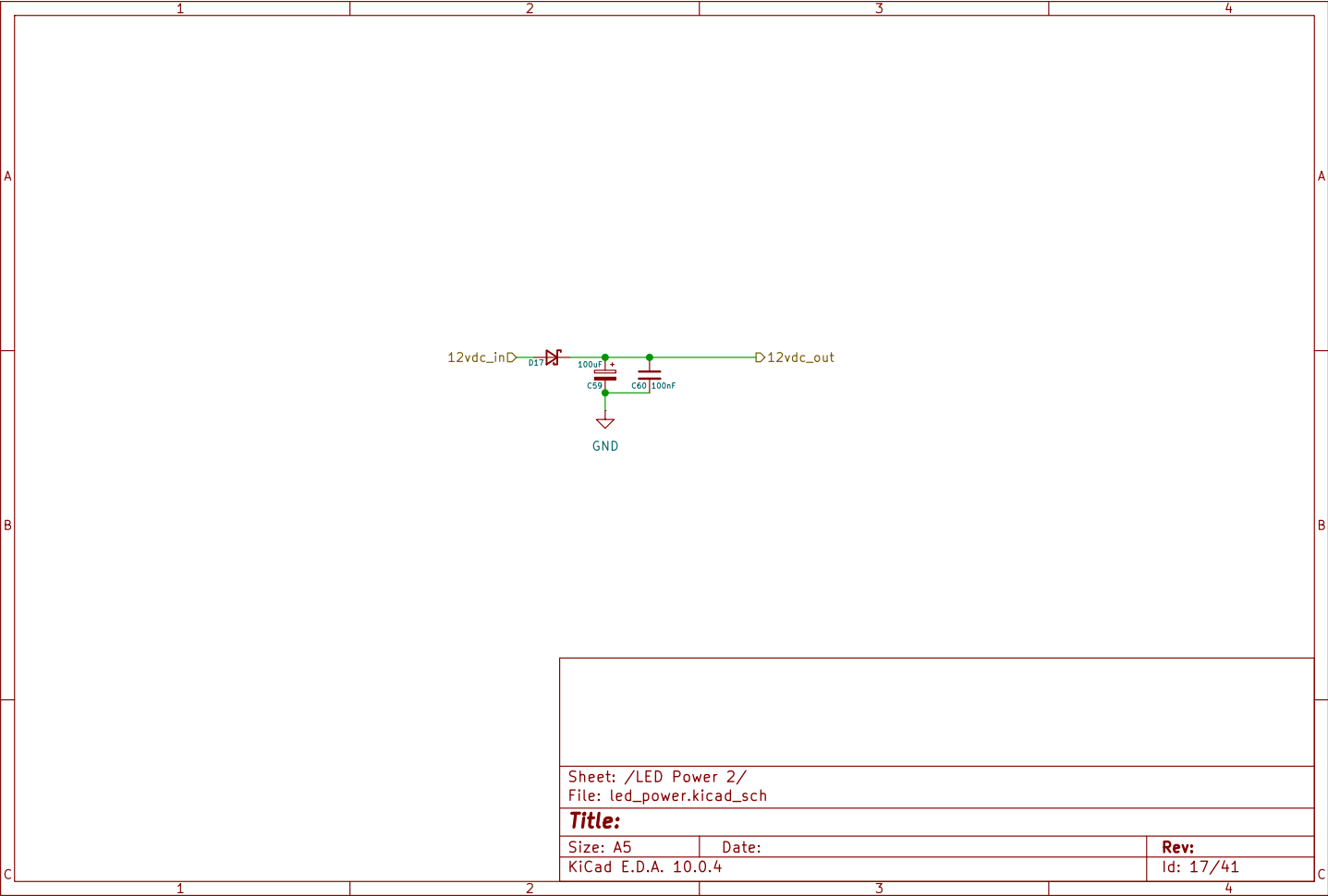
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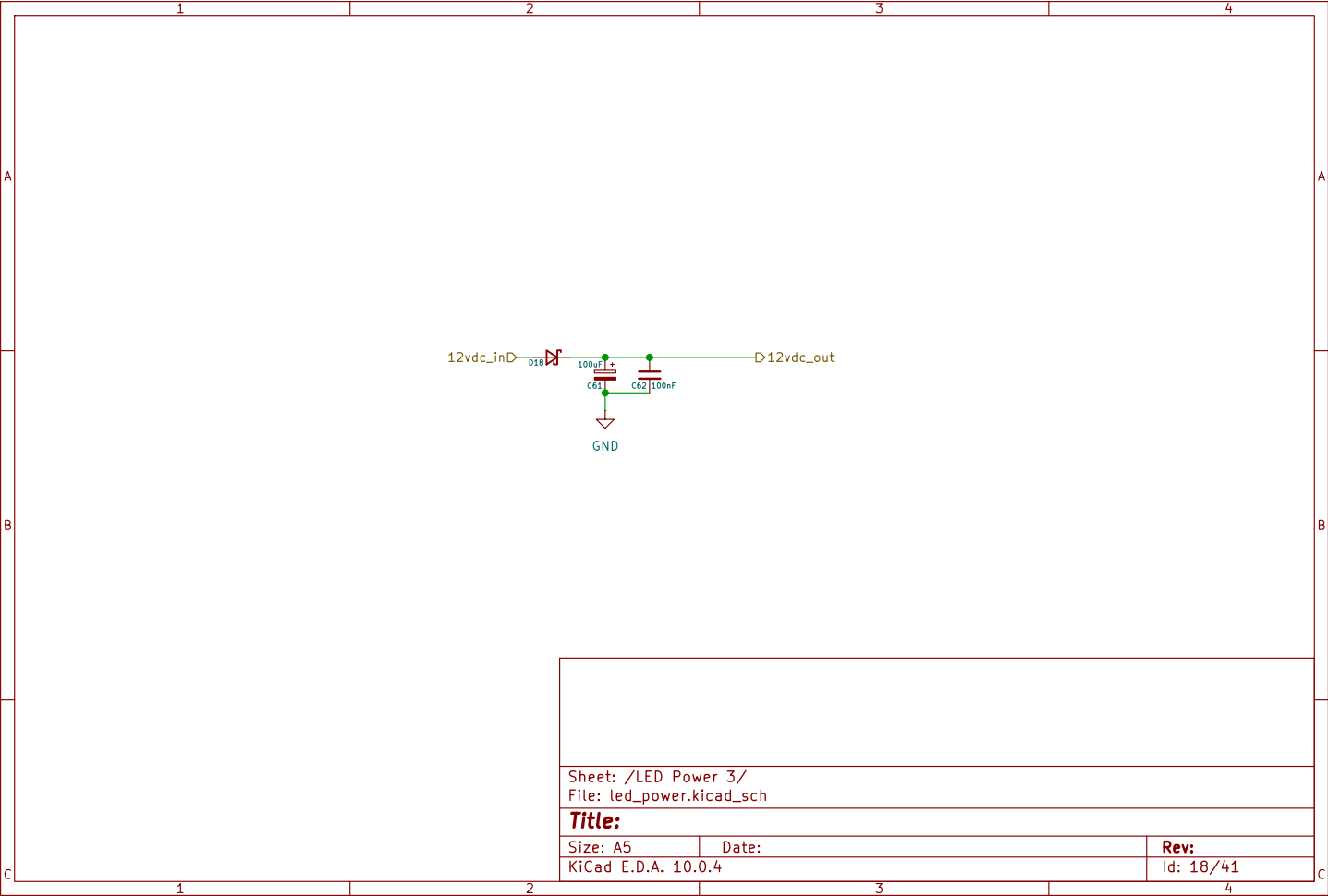
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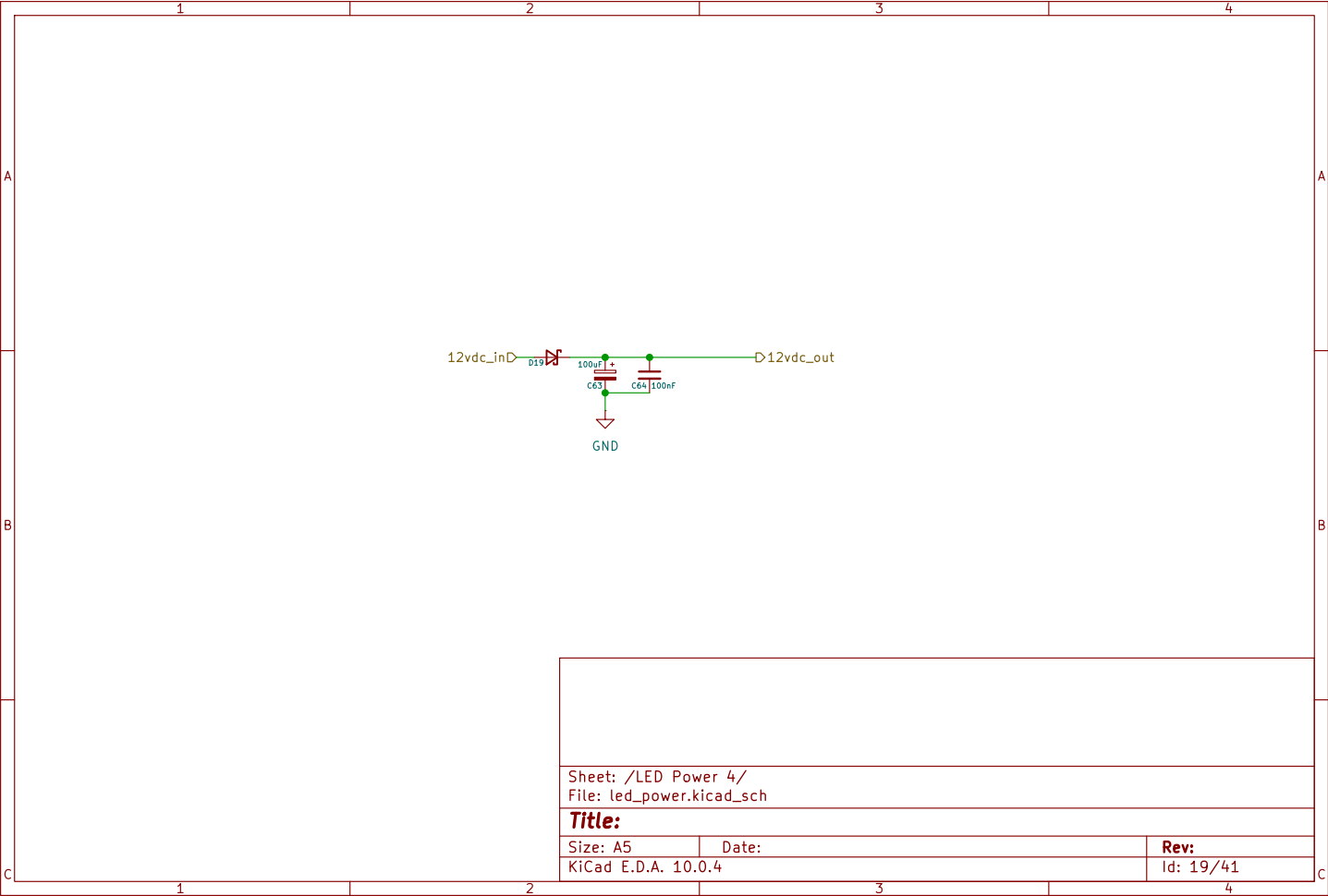
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Id: 15/41









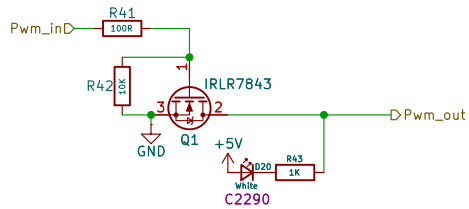
Sheet: /LED Power 4/
File: led_power.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 19/41



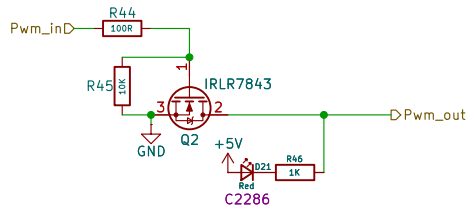
Sheet: /LED Channel 1/
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Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 20/41



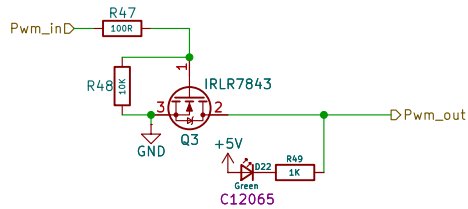
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Title:

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Date:

Rev:
Id: 21/41



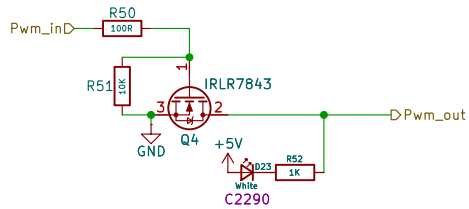
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Title:

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Date:

Rev:
Id: 23/41



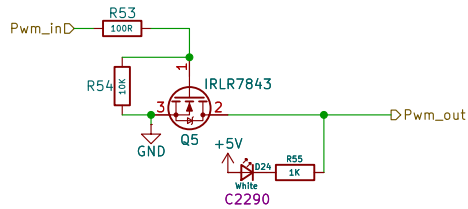
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Title:

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Date:

Rev:
Id: 24/41



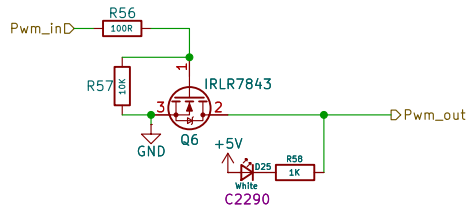
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Title:

Size: A5
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Date:

Rev:
Id: 25/41



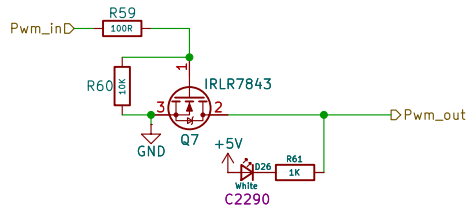
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Title:

Size: A5
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Date:

Rev:
Id: 26/41



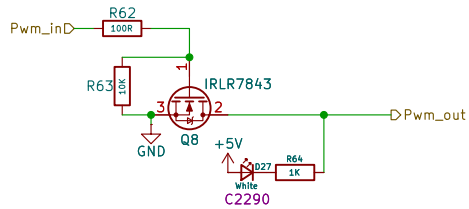
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File: led_channel.kicad_sch

Title:

Size: A5
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 27/41



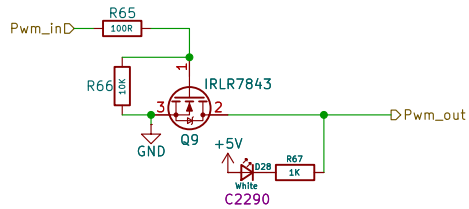
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Title:

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Date:

Rev:
Id: 28/41



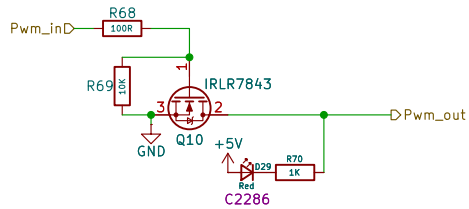
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Title:

Size: A5
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Date:

Rev:
Id: 29/41



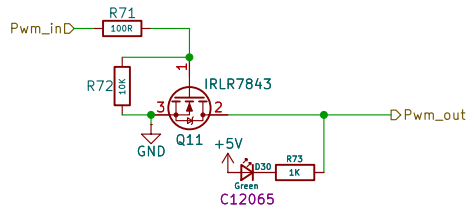
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Title:

Size: A5
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Date:

Rev:
Id: 30/41



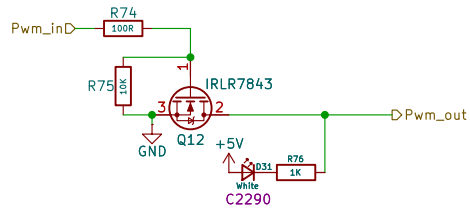
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Title:

Size: A5
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Date:

Rev:
Id: 31/41



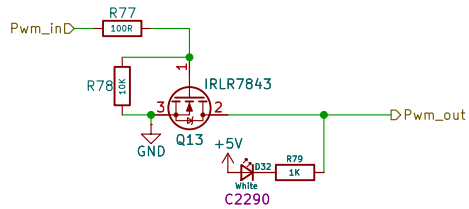
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Title:

Size: A5
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Date:

Rev:
Id: 32/41



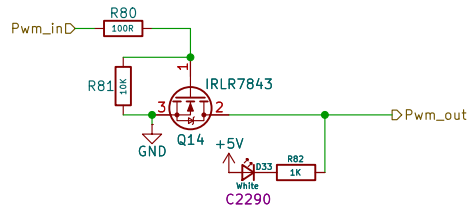
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Title:

Size: A5
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Date:

Rev:
Id: 33/41



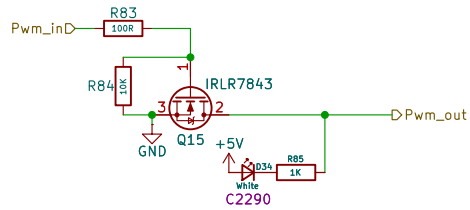
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Title:

Size: A5
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Date:

Rev:
Id: 34/41



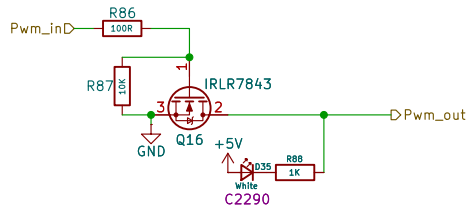
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Title:

Size: A5
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Date:

Rev:
Id: 35/41



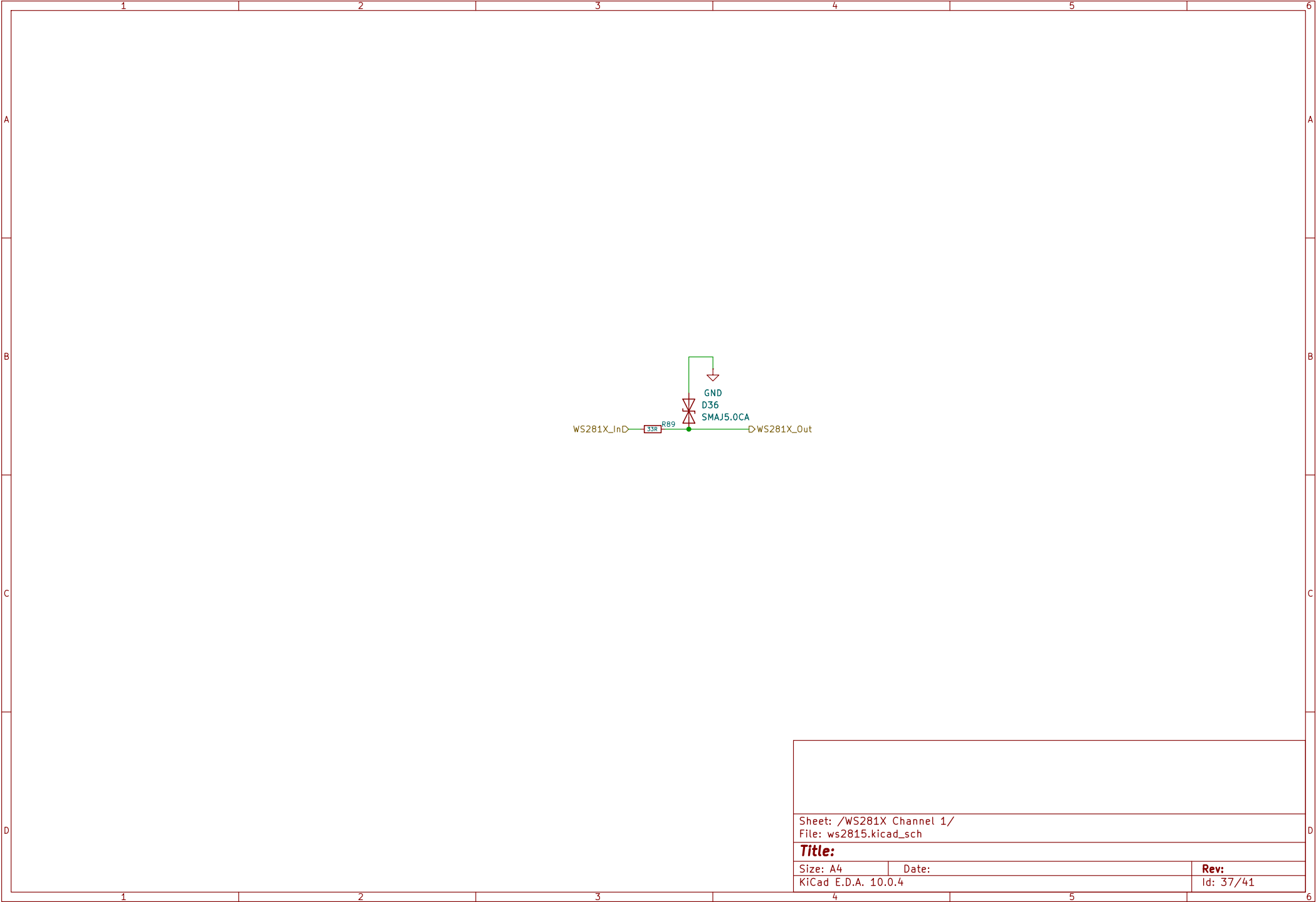
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File: led_channel.kicad_sch

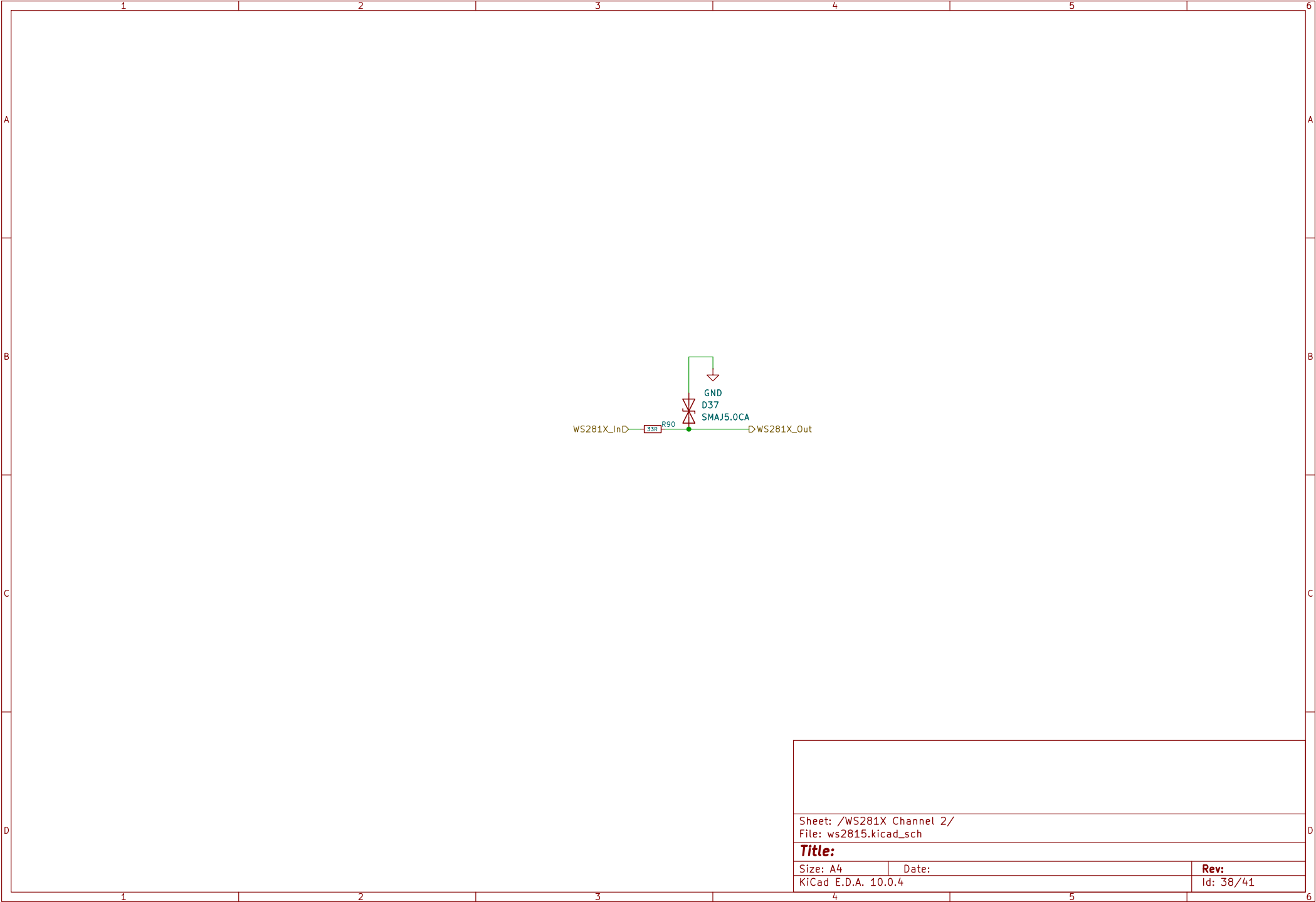
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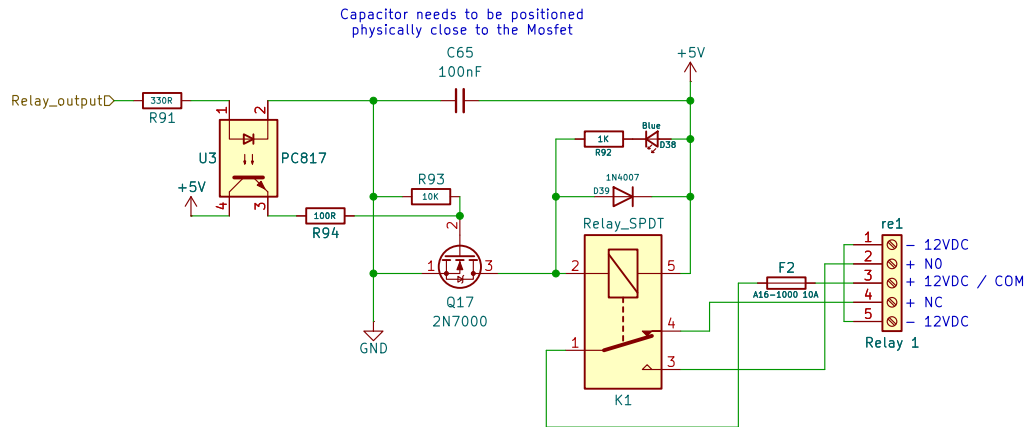
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Date:

Rev:
Id: 36/41







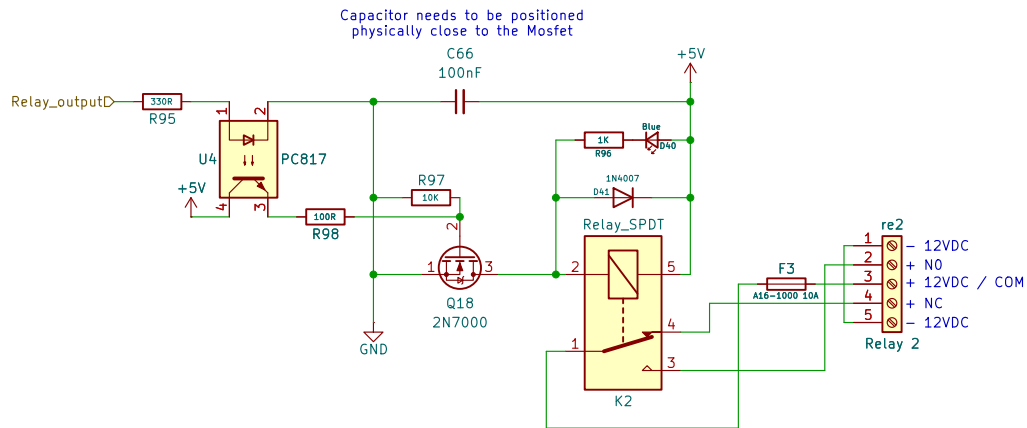
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Title:

Size: A5
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Date:

Rev:
Id: 39/41



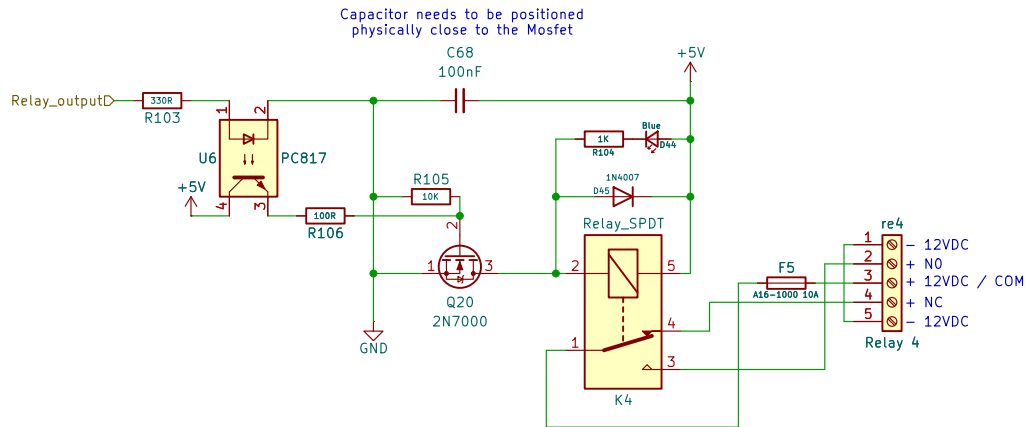
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Title:

Size: A5
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Date:

Rev:
Id: 40/41



Sheet: /Relay Channel 4/
File: relay_channel.kicad_sch

Title:

Size: A5
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Date:

Rev:
Id: 42/41